THE SECTIONED PICARD STACK IS FIBRED IN SETOIDS WITH KERNEL ISOMORPHISM CLASSES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-pic (canonical tag 0D25). cited at statement line 3192. Standing situation with a section.

Source reference. Original definition 0D25: sources/stacks-chow/tags.txt, source lines 16370–16370. Cited at statement line 3192: standing situation.

Lemma 0.2. *In Situation 0D25 let $\sigma : B \to X$ be a section. If $\mathcal{O}_T \to f_{T,*}\mathcal{O}_{X_T}$ is an isomorphism for all T over B, then $\mathcal{P}ic_{X/B,\sigma} \to (Sch/S)_{fppf}$ is fibred in setoids with set of isomorphism classes over T given by*

$$\coprod_{h:T\to B} \mathrm{Ker}(\sigma_T^* : \mathrm{Pic}(X \times_{B,h} T) \to \mathrm{Pic}(T))$$

Proof. If $\xi = (T, h, \mathcal{L}, \alpha)$ is an object of $\mathcal{P}ic_{X/B,\sigma}$ over T, then an automorphism φ of ξ is given by multiplication with an invertible global section u of the structure sheaf of X_T such that moreover $\sigma_T^* u = 1$. Then $u = 1$ by our assumption that $\mathcal{O}_T \to f_{T,*}\mathcal{O}_{X_T}$ is an isomorphism. Hence $\mathcal{P}ic_{X/B,\sigma}$ is fibred in setoids over $(Sch/S)_{fppf}$. Given T and $h : T \to B$ the set of isomorphism classes of pairs $(\mathcal{L}, \alpha)$ is the same as the set of isomorphism classes of $\mathcal{L}$ with $\sigma_T^*\mathcal{L} \cong \mathcal{O}_T$ (isomorphism not specified). This is clear because any two choices of α differ by a global unit on T and this is the same thing as a global unit on X_T. □

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